

Diabetic Retinopathy Grading from Fundus Photographs: An Ordinal Classification Approach using Deep Learning

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Abstract

1 Introduction

Diabetic Retinopathy (DR) is a complication of diabetes and a leading cause of preventable blindness. It is graded on a five-point scale, from no retinopathy (Grade 0), through mild, moderate, and severe non-proliferative DR, to proliferative DR (Grade 4). Early stages are asymptomatic, so screening relies on colour fundus photography, where clinicians look for microaneurysms, haemorrhages, exudates, and neovascularisation. Manual grading is slow, requires specialist expertise, and varies between graders, which has motivated automated deep-learning systems that grade fundus images directly.

For this work we use the publicly available **APTOS 2019 Blindness Detection** dataset (Kaggle), containing 3,662 high-resolution colour fundus photographs from rural India, labelled by clinicians on the five-class International Clinical Diabetic Retinopathy (ICDR) severity scale (0–4).

2 Problem Statement

The task of DR grading from fundus photographs is fundamentally an *ordinal* five-class classification problem and the cost of misclassification grows with the distance between the predicted and true grade. Standard image classification models that treat the grades as independent categorical labels therefore discard valuable information and tend to produce clinically implausible errors. The problem is further complicated by a severe class imbalance, since the majority of patients exhibit no or mild disease, leaving the higher-severity grades drastically under-represented in available datasets. The objective of this work is to design a convolutional neural network that perform ordinal five-class grading upon classification, mitigates class imbalance, and produces interpretable, clinically aligned predictions evaluated through the Quadratic Weighted Kappa (QWK) metric and grad-CAM analysis to identify which regions of the image are most important for each prediction.

2.1 Baseline Model

We approached the problem the way any standard image classification task would be approached: with an ImageNet-pretrained CNN, a linear head re-mapped to the five DR grades, and a vanilla cross-entropy loss. Three backbones of progressively increasing capacity were chosen so that we could observe how depth and architectural sophistication interact with our small (~ 3.6 k-image) and heavily imbalanced training set. The first model, used as our reference baseline, is **ResNet-18** (~ 11.2 M parameters)—a residual network deep enough to learn rich

features but light enough to train end-to-end on a single 4 GB consumer GPU. The second is **ResNet-50** (~ 25 M parameters), chosen to test whether simply going deeper improves performance under identical training conditions. The third is **EfficientNet-B0** (~ 5 M parameters), selected as a more parameter-efficient, compound-scaled architecture that often dominates the medical imaging literature for fundus tasks. Across all three runs the optimisation recipe was kept deliberately fixed—Adam optimiser at $\text{lr} = 10^{-4}$, 224×224 input resolution, 10 training epochs, an 80/20 stratified train-validation split, and unweighted **categorical cross-entropy** as the loss function—so that any differences in performance can be traced back to the architecture rather than the optimisation strategy.

2.2 Baseline Results

The ResNet-18 baseline achieved a best validation Quadratic Weighted Kappa of **QWK** = 0.8819 and accuracy of **83.49%**, meeting standard performance benchmarks. However, the core finding from the baseline experiments reveals a critical failure: **severe overfitting and class imbalance**. The training and validation curves (Figure 1) clearly show the divergence between training and validation loss over 10 epochs, indicating the model is memorising patterns rather than learning generalizable features.

Table 1: ResNet-18 Baseline Classification Report

Grade	Precision	Recall	F1-Score	Support
Grade 0	0.94	0.96	0.95	361
Grade 1	0.50	0.04	0.07	74
Grade 2	0.57	0.94	0.71	200
Grade 3	0.64	0.18	0.28	39
Grade 4	0.72	0.22	0.34	59
Accuracy	0.76			733

The per-class classification performance (Table 1) exposes the clinical inadequacy of this approach. While Grade 0 (healthy) is recognised almost perfectly (F1-score = 0.95), the minority and clinically critical grades severely degrade: Grade 3 and Grade 4 show F1-scores of only 0.28 and 0.34 respectively. The model fails where it matters most—on advanced disease stages. This pattern confirms that standard categorical cross-entropy loss and naive training are fundamentally unsuited to this ordinal, imbalanced medical imaging task.

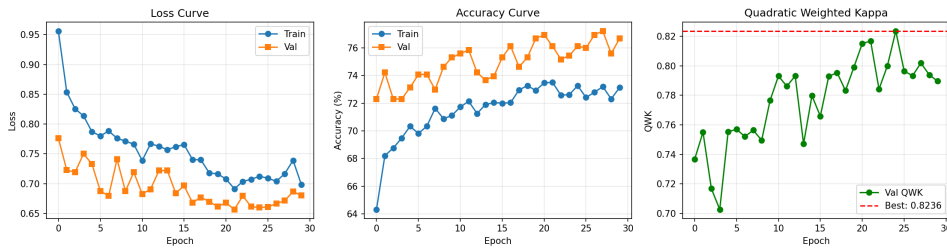


Figure 1: Training and validation loss curves over 10 epochs, showing characteristic overfitting behaviour.

2.3 Defining Objectives

Through baseline experiments, it is evident that *any* typical image classification CNN, when transplanted directly onto APTOS-2019, fails in the same predictable way: it overfits a small, heavily class-imbalanced medical dataset differ sharply from the natural-image distribution the

backbone was pretrained on. The objectives of this project therefore reorient away from chasing yet another architecture and towards directly attacking the root causes:

1. **Reduce overfitting** through processes like data augmentation, regularisation (dropout, weight decay), and stronger normalisation of the train/validation curves.
2. **Fine-tune ResNet and find an appropriate loss function** that better capture the ordinal nature of the problem, such as focal loss and CORAL loss that explicitly models the rank order of the classes.
3. **Explore alternatives to flat softmax classification** that respect the ordinal structure and class imbalance of DR severity, including weighted cross-entropy,

3 Literature Review

3.1 Model Selection

DR classification has largely tracked the broader ImageNet backbone progression. Gulshan *et al.* [1] set the template with an **Inception-v3**-based binary referable-vs-non-referable classifier that reached ophthalmologist-level sensitivity on 128k images, after which work on five-class severity grading shifted to deeper residual networks (**ResNet-50**, **DenseNet-121**) and compound-scaled **EfficientNet** variants. Across these systems the classification head is uniformly a flat softmax over Grades 0–4, which discards the ordinal structure and penalises a Grade-4-as-Grade-0 confusion no more than a Grade-4-as-Grade-3 one—an assumption consistent with the high inter-grader disagreement reported on the same images by human ophthalmologists [2].

For our setting—a ~ 3.6 k-image training set on a single 4 GB consumer GPU, with ordinality to be supplied by the loss rather than the backbone—the relevant axis is not raw capacity but the smallest backbone that still gives useful ImageNet-pretrained features without overfitting. **ResNet-50** (~ 25 M parameters) brings roughly $2.2\times$ the parameters of ResNet-18 and, at this data scale, simply enlarges the overfitting surface; **EfficientNet-B0** (~ 5 M parameters) is parameter-efficient on paper but its compound-scaling advantage requires more data than we have to be realised. **ResNet-18** (~ 11.2 M parameters) sits at the sweet spot—deep enough to learn the multi-scale lesion textures that distinguish DR severity grades, shallow enough to fine-tune end-to-end in memory, and well-matched to the data regime so that gains from regularisation and ordinal-aware losses are not masked by capacity-driven overfitting. We therefore adopt **ResNet-18** as the backbone and direct the remaining effort towards the classification head, the loss function, and the preprocessing pipeline.

3.2 Preprocessing

Preprocessing is not a cosmetic step in medical imaging, it is the single largest lever between a model that learns lesion-level pathology and one that latches onto background artefacts. Raw fundus photographs vary enormously in illumination, contrast, white-balance, the size of the visible retinal disc, and the amount of black border surrounding it; without standardisation, the network is forced to spend representational capacity on these nuisance factors instead of on disease cues [13, 14]. A consistent finding in the medical CNN literature is that, in the absence of preprocessing, gradient-based attribution maps drift onto clinically irrelevant regions (background, large vessels, image borders), whereas appropriate preprocessing concentrates attention on the diagnostic structures themselves [13].

Within our pipeline, preprocessing is built around four mutually reinforcing components. **Geometric normalisation** resizes every image to a fixed 224×224 resolution and crops the circular field of view, removing the black corners that otherwise contribute uninformative gradients.

Photometric normalisation subtracts the per-channel ImageNet mean (0.485, 0.456, 0.406) and divides by its standard deviation (0.229, 0.224, 0.225), ensuring that the feature statistics fed to the pretrained ResNet match those it was trained on, without this step, transfer learning is markedly less effective. **Data augmentation** (horizontal/vertical flips, small rotations, brightness-contrast jitter, CLAHE contrast enhancement) synthetically expands the effective training distribution and is one of the most powerful regularisers known for limited medical datasets [12, 7]. **Regularisation hyperparameters** that complement preprocessing, most notably *dropout*, which randomly disables units during training to prevent co-adaptation [10], and *weight decay*, which penalises large parameter norms, directly address the overfitting we observed in the baseline. Closely related are the choice of *weight initialisation* [11], the optimiser learning rate, and the use of pretrained ImageNet weights.

3.3 Loss Function Definition

Initially, the most obvious choice for a loss function is the standard **categorical cross-entropy**, which treats the five DR grades as independent classes. However, this approach discards the ordinal structure of the problem and fails to account for class imbalance, later introduced was **weighted cross-entropy** in [6], which assigns higher loss weights to the minority classes (Grades 3 and 4) to mitigate imbalance. **Focal loss** [5] further refines this by down-weighting easy examples and focusing the model’s attention on hard, misclassified samples, which is particularly beneficial in imbalanced settings. This brings our class imbalance down to a manageable level, but it still treats the classes as independent. **Ordinal regression losses** such as CORAL [9] explicitly model the rank order of the classes by decomposing the problem into a series of binary classification tasks, ensuring that the model learns to respect the natural ordering of DR severity. This handles the severity of misclassification, for example, misclassifying Grade 4 as Grade 0 is penalised more than misclassifying it as Grade 3. This is a critical improvement over standard cross-entropy, which treats all misclassifications as equally bad.

3.4 Evaluation Metric

Plain accuracy is poorly suited to a five-class, severely imbalanced, ordinal task: predicting Grade 0 for every image already attains close to 50% accuracy on APTOS-2019 without learning anything clinically useful. We therefore adopt the **Quadratic Weighted Kappa (QWK)** as the primary metric, following Cohen’s weighted-kappa framework [15], which rewards agreement between predicted and true grades and penalises disagreement by the *squared* distance between them, so a Grade-4-as-Grade-0 error is sixteen times costlier than an adjacent-grade error. QWK is the official scoring metric of both the Kaggle *Diabetic Retinopathy Detection* (2015) and *APTOS-2019 Blindness Detection* competitions, and de la Torre *et al.* [16] have shown that it doubles as a differentiable training loss for DR grading.

We supplement QWK with overall **accuracy** (for comparability with prior reports, while remembering its insensitivity to imbalance [6]), **macro-F1** (equally-weighted per-class F1, which surfaces collapses on minority Grades 3 and 4), and the full **confusion matrix** (which reveals *which* grades are being conflated and is what exposed the baseline’s collapse onto Grades 0 and 2).

4 Methodology

As discussed previously, the baseline model’s performance is significantly constrained by two major limitations in the dataset and problem structure: severe class imbalance and the ordinal nature of diabetic retinopathy severity grades. The dataset is heavily skewed toward lower-grade samples, causing the model to bias predictions toward majority classes while underperforming on minority classes representing more severe disease stages. In addition, diabetic retinopathy

classification is inherently ordinal, meaning that misclassifications should not all be treated equally. For example, predicting a Grade 4 image as Grade 3 is considerably less severe than predicting it as Grade 0, yet standard Cross Entropy Loss penalizes both errors similarly because it treats classes as independent categories rather than ordered progression levels. This limits the model’s ability to learn the structured severity relationship between grades.

To address these shortcomings, we propose the use of a hybrid loss function that combines Focal Loss and CORAL (Consistent Rank Logits) Loss. Focal Loss is specifically designed to mitigate class imbalance by dynamically down-weighting easy, high-confidence predictions and assigning greater emphasis to hard or misclassified examples. This will ensure that minority and difficult samples contribute more significantly during backpropagation, improving sensitivity to underrepresented disease stages. CORAL Loss, on the other hand, is designed for ordinal classification problems by transforming multi-class prediction into a series of ordered binary threshold tasks. Instead of treating grades as unrelated labels, CORAL will learn rank-consistent decision boundaries that explicitly preserve severity progression, allowing the model to better understand the ordered relationship between diabetic retinopathy grades. By integrating Focal weighting into the CORAL framework, the proposed loss function will simultaneously address both imbalance and ordinality, enabling the model to prioritize difficult minority samples while preserving clinically meaningful grading structure. The following is the loss function we formulated:

$$\mathcal{L}_{\text{Focal-CORAL}} = - \sum_{i=1}^N \sum_{k=1}^{K-1} \alpha_k (1 - p_{t,i}^{(k)})^\gamma \log(p_{t,i}^{(k)})$$

$$p_{t,i}^{(k)} = \begin{cases} p_i^{(k)}, & \text{if } y_i^{(k)} = 1 \\ 1 - p_i^{(k)}, & \text{if } y_i^{(k)} = 0 \end{cases}$$

$$p_i^{(k)} = \sigma(z_i^{(k)})$$

$$y_i^{(k)} = \mathbb{I}(y_i > k), \quad k = 1, 2, \dots, K - 1$$

In addition to loss function improvements, extensive preprocessing will be applied to enhance image quality and reduce irrelevant variability in retinal fundus images. Medical retinal images often suffer from inconsistent illumination, poor contrast, noise, and imaging artifacts due to differences in acquisition devices and conditions. To standardize the dataset, Ben Graham preprocessing will be employed to normalize illumination, suppress background variation, and enhance retinal structures by subtracting local average intensity. This will improve visibility of important pathological features such as blood vessels, microaneurysms, exudates, and hemorrhages. Furthermore, Contrast Limited Adaptive Histogram Equalization (CLAHE) will be applied to improve local contrast, particularly in darker or low-quality regions, making subtle lesions more distinguishable while preventing excessive noise amplification. These preprocessing techniques will collectively improve feature clarity, reduce inter-image variability, and help the model focus on clinically relevant patterns rather than acquisition-related inconsistencies.

To further address class imbalance, data augmentation strategies will also be applied more aggressively to minority classes. Transformations such as rotation, flipping, zooming, brightness adjustments, and cropping will increase sample diversity while preserving pathological relevance. This will not only improve representation of under-sampled severity grades but also enhance model generalization by exposing it to a broader range of retinal variations. Together, the combination of hybrid ordinal-imbalance-aware loss functions and robust preprocessing will create a more clinically informed and technically optimized framework for diabetic retinopathy classification, with the goal of improving predictive accuracy across all severity levels, especially in minority and early-detection classes where accurate diagnosis is most critical.

Building upon the baseline established using ResNet-18, which was initially trained without advanced preprocessing or loss function enhancements, we plan to further investigate whether alternative architectures can provide superior performance when combined with the proposed methodological improvements. Specifically, we will evaluate DenseNet and EfficientNet alongside the improved ResNet-18 framework after integrating Ben Graham preprocessing, CLAHE-based contrast enhancement, data augmentation, and the combined Focal-CORAL loss function. Each architecture will be trained and evaluated under the same enhanced preprocessing and loss function framework to ensure fair comparison. Performance will be assessed using metrics such as accuracy, F1-score, recall, and sensitivity across all diabetic retinopathy severity grades, with particular attention to minority and severe classes. In addition to predictive performance, model complexity and parameter count will also be analyzed. This comparative strategy will allow us to quantify not only the gains achieved through methodological improvements over the original ResNet-18 baseline, but also determine which architecture provides the most effective balance between classification performance and practical deployment considerations.

5 Results

6 Future Directions

7 Conclusion

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